Lab 7: Fuzzing for Input Validation Bugs

Project: L6_L7_20_food_delivery (Food Delivery)
Tools: Python 3.14.3, pytest 9.2.0, Hypothesis 6.170.15

Step 2: Output of the buggy application (python3 food_delivery.py)

```
(venv) pravith@Praviths-MacBook-Air-2 Lab 7 % python3 food_delivery.py
==== Running parse_order ====
Enter items as name:qty CSV (e.g. pizza:2,burger:1): pizza:2,burger:1
Order: {'pizza': 2, 'burger': 1}

==== Running delivery_eta ====
Enter distance in km (e.g. 4.5): 4.5
ETA in Minutes: 33

==== Running coupon_discount ====
Enter a coupon code (e.g. SAVE50): SAVE50
Discount: 50
(venv) pravith@Praviths-MacBook-Air-2 Lab 7 %
```

Step 4: Fuzz tests failing on the buggy code (python -m pytest test_food_delivery.py -v)

Hypothesis found 3 failures in parse_order (inputs ':', '', None), 2 in delivery_eta ('', None) and 3 in coupon_discount ('SAVE', '', None).

```
(venv) pravith@Praviths-MacBook-Air-2 Lab 7 % python -m pytest test_food_delivery.py -v
============================= test session starts ==============================
platform darwin -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- /Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/path/to/venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_food_delivery.py::test_parse_order_no_crash FAILED                  [ 33%]
test_food_delivery.py::test_delivery_eta_no_crash FAILED                 [ 66%]
test_food_delivery.py::test_coupon_discount_no_crash FAILED              [100%]

=================================== FAILURES ===================================
_____________________________ test_parse_order_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 8, in test_parse_order_no_crash
  |     def test_parse_order_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 10, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 5, in parse_order
    |     order[name.strip()] = int(qty)
    |                           ~~~^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 12, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with input {s!r}: {e}"
    | AssertionError: parse_order crashed with input ':': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 10, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 4, in parse_order
    |     name, qty = item.split(':')
    |     ^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
```

```
    |     assert False, f"parse_order crashed with input {s!r}: {e}"
    | AssertionError: parse_order crashed with input ':': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 10, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 4, in parse_order
    |     name, qty = item.split(':')
    |     ^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 12, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with input {s!r}: {e}"
    | AssertionError: parse_order crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 10, in test_parse_order_no_crash
    |     parse_order(s)
    |     ~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 3, in parse_order
    |     for item in items_csv.split(','):
    |                 ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 12, in test_parse_order_no_crash
    |     assert False, f"parse_order crashed with input {s!r}: {e}"
    | AssertionError: parse_order crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_order_no_crash(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
_____________________________________________________________ test_delivery_eta_no_crash _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 17, in test_delivery_eta_no_crash
```

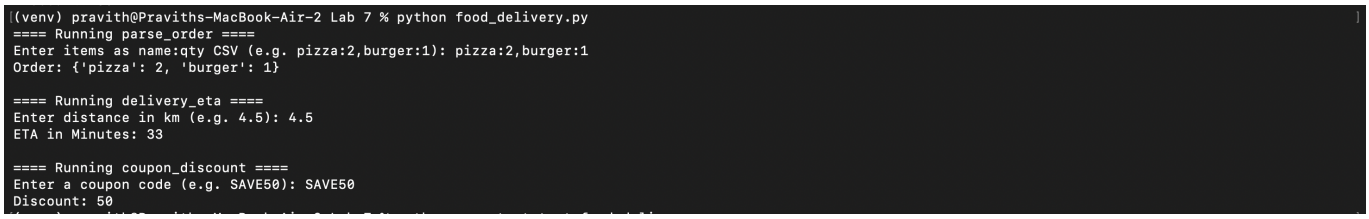
```
    | )
    +------------------------------------
_____________________________________________________________ test_delivery_eta_no_crash _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 17, in test_delivery_eta_no_crash
  |     def test_delivery_eta_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 19, in test_delivery_eta_no_crash
    |     delivery_eta(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 9, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    |                     ~~~~~^^^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 21, in test_delivery_eta_no_crash
    |     assert False, f"delivery_eta crashed with input {s!r}: {e}"
    | AssertionError: delivery_eta crashed with input '': could not convert string to float: ''
    | assert False
    | Failing test case: test_delivery_eta_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py:20
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 19, in test_delivery_eta_no_crash
    |     delivery_eta(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 9, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    |                     ~~~~~^^^^^^^^^^^^^^
    | TypeError: float() argument must be a string or a real number, not 'NoneType'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 21, in test_delivery_eta_no_crash
    |     assert False, f"delivery_eta crashed with input {s!r}: {e}"
    | AssertionError: delivery_eta crashed with input None: float() argument must be a string or a real number, not 'NoneType'
    | assert False
    | Failing test case: test_delivery_eta_no_crash(
    |     s=None,
```

```
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 19, in test_delivery_eta_no_crash
    |     delivery_eta(s)
    |     ~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 9, in delivery_eta
    |     return int(15 + float(distance_km) * 4)
    |                     ~~~~~^^^^^^^^^^^^^
    | TypeError: float() argument must be a string or a real number, not 'NoneType'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 21, in test_delivery_eta_no_crash
    |     assert False, f"delivery_eta crashed with input {s!r}: {e}"
    | AssertionError: delivery_eta crashed with input None: float() argument must be a string or a real number, not 'NoneType'
    | assert False
    | Failing test case: test_delivery_eta_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         /Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py:20
    +------------------------------------
_____________________________________________________________ test_coupon_discount_no_crash _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 26, in test_coupon_discount_no_crash
  |     def test_coupon_discount_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 30, in test_coupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with input {s!r}: {e}"
    | AssertionError: coupon_discount crashed with input 'SAVE': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s='SAVE',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

```
_____________________________________________________________ test_coupon_discount_no_crash _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 26, in test_coupon_discount_no_crash
  |     def test_coupon_discount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 30, in test_coupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with input {s!r}: {e}"
    | AssertionError: coupon_discount crashed with input 'SAVE': invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s='SAVE',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 30, in test_coupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with input {s!r}: {e}"
    | AssertionError: coupon_discount crashed with input '': list index out of range
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
```

```
    | Failing test case: test_coupon_discount_no_crash(
    |     s='SAVE',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 30, in test_coupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with input {s!r}: {e}"
    | AssertionError: coupon_discount crashed with input '': list index out of range
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 28, in test_coupon_discount_no_crash
    |     coupon_discount(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/food_delivery.py", line 12, in coupon_discount
    |     return int(code.upper().split('SAVE')[1])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'upper'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/pravith/Desktop/L6_L7_20_food_delivery/Lab 7/test_food_delivery.py", line 30, in test_coupon_discount_no_crash
    |     assert False, f"coupon_discount crashed with input {s!r}: {e}"
    | AssertionError: coupon_discount crashed with input None: 'NoneType' object has no attribute 'upper'
    | assert False
    | Failing test case: test_coupon_discount_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_food_delivery.py::test_parse_order_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_food_delivery.py::test_delivery_eta_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_food_delivery.py::test_coupon_discount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 0.94s ===============================
(venv) pravith@Praviths-MacBook-Air-2 Lab 7 %
```

Step 5: Output of the fixed application (python food_delivery.py)



Step 6: Fuzz tests re-run on the fixed code (python -m pytest test_food_delivery.py -v)



All 3 tests passed after hardening the functions.